

Multimodal Optimistic and Opportunistic E-Commerce Fashion Product Recommendation

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Abstract—In the dynamic realm of e-commerce, delivering efficient and accurate product recommendations is essential for enhancing user experience and increasing conversion rates. This paper introduces a multimodal fashion product recommendation system that leverages both text and image inputs for improved relevance and personalization. Visual features are extracted using the VGG19 model, while textual features are processed using LSTM networks. To manage high-dimensional data, dimensionality reduction techniques like PCA and UMAP are employed, followed by similarity computation using KNN, cosine similarity, and Annoy. The proposed system achieves a notable accuracy of 93.2 with PCA and Annoy, significantly outperforming the benchmark accuracy of 89.3 from the base paper. Moreover, the system is almost 20x faster, enabling real-time performance in large-scale e-commerce environments in comparison to other recommendation techniques. By combining multimodal inputs and advanced algorithms, this work establishes a scalable, dynamic approach to intelligent product recommendations.

Index Terms—Multimodal, Annoy, UMAP, PCA, LSTM

I. INTRODUCTION

The rapid expansion of e-commerce has reshaped the global retail landscape, enabling consumers to browse and purchase products online with unparalleled ease and convenience. However, with the vast variety of products available, finding the right items that meet individual preferences can be overwhelming. Traditional product search systems in e-commerce platforms have relied predominantly on either textual descriptions or images to recommend products. Text-based systems focus on product titles, categories, and descriptions, whereas image-based systems extract features from visual content, such as colors, shapes, and patterns. While both approaches have their strengths, they often fall short in fully capturing the diverse preferences and intricate details that users seek in fashion products.

In recent years, the integration of multimodal data, combining both text and images, has emerged as an effective strategy to address these limitations. By leveraging the complementary strengths of text and image data, multimodal recommendation systems offer a richer, more accurate representation of both products and user intent. This is particularly important in the fashion domain, where visual appeal is just as important as textual descriptions in influencing consumer decisions. However,

combining these two modalities introduces challenges related to feature extraction, data complexity, and computational efficiency, particularly with large-scale datasets.

In this paper, we propose a novel multimodal recommendation system for fashion products, specifically designed to overcome these challenges. The system extracts visual features using the VGG19 convolutional neural network, a deep learning model known for its effectiveness in image recognition tasks. For textual feature extraction, we employ Long Short-Term Memory (LSTM) networks, which are capable of capturing sequential dependencies and contextual information in product descriptions. To address the high-dimensional nature of the extracted features, we apply dimensionality reduction techniques such as Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP).

To assess the similarity between products, we utilize various techniques including K-Nearest Neighbors (KNN), cosine similarity, and the Approximate Nearest Neighbor (Annoy) algorithm. The proposed system significantly improves recommendation accuracy, achieving an impressive 93.2 accuracy rate, while being 20 times faster than traditional approaches. This performance enhancement is crucial for real-time applications in e-commerce, where speed and relevance are key to improving user experience and increasing conversion rates.

The findings from this research demonstrate that by combining text and image features and optimizing the computational efficiency of the recommendation process, multimodal systems can offer highly personalized, scalable, and real-time product recommendations. This paper presents a comprehensive analysis of the methodology, experimental results, and the potential impact of such systems in the evolving e-commerce landscape.

II. RELATED WORKS

The integration of multiple modalities, particularly text and image, in product recommendation systems has been a growing area of research in e-commerce. Traditional recommendation systems have largely been based on either textual or visual data, but recent studies have demonstrated the benefits of combining these two modalities for more accurate and personalized recommendations.

A. Vision-and-Language Transformers

Multimodal retrieval in the fashion domain has increasingly leveraged Vision-and-Language (V+L) Transformers, which combine text and image processing to improve retrieval accuracy. Early models like FashionBERT and KaleidoBERT rely on pre-trained ResNet networks for image feature extraction, but this approach can result in high memory usage and computational costs, limiting scalability in large-scale e-commerce applications. Newer models, such as MVLT, bypass the need for pre-processed feature vectors and instead work directly with raw image patches, applying more efficient training techniques like masked image reconstruction tasks. However, these approaches often overlook the importance of optimizing similarity measurements between images and text.

Moreover, traditional transformer-based models require simultaneous processing of text and image inputs, making them inefficient for large datasets and hindering the use of multi-dimensional indices, such as PCA and Product Quantization. As a result, there has been a shift toward dual encoding, where image and text are processed separately and aligned later for more efficient retrieval.

Our approach improves upon these methods by using a V+L transformer that bypasses ResNet, directly processes image patches, and integrates efficient similarity computations for better feature representation and faster retrieval. This results in a more scalable and efficient framework for real-time fashion product recommendation systems.

B. Advancements in Multimodal Retrieval Using Deep Metric Learning

In multimodal retrieval, the ability to measure the similarity between text and image features accurately is crucial. Early works employed methods like triplet loss to map both text and image features into a shared latent space, where similarity is measured using Euclidean distance. Further improvements in similarity-based models, such as VSE++, introduced techniques to focus on more informative samples during training, significantly improving retrieval performance. Models like SCAN and PFAN incorporated attention mechanisms to better model the interactions between image and text features, enhancing the quality of multimodal embeddings.

In fashion retrieval, contrastive pretraining techniques are frequently used, where separate encoders are employed for text and image data, aligning them into a shared latent space. While effective, these methods tend to be computationally expensive and require substantial resources for training, limiting their scalability.

Our method distinguishes itself by combining self-supervised learning with efficient similarity computations to create multimodal representations. We apply PCA and UMAP for dimensionality reduction, followed by Annoy for fast and efficient similarity computation. This combination ensures high performance and scalability, enabling real-time, large-scale multimodal recommendation tasks, particularly in the fashion e-commerce sector.

C. Dimensionality Reduction and Similarity Measurement

To manage the high-dimensional data produced by multimodal models, dimensionality reduction techniques like Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are widely used. These methods reduce feature complexity while retaining important data characteristics, leading to more efficient computations. PCA is commonly used to reduce feature space by identifying the principal components that capture the maximum variance. UMAP, on the other hand, preserves both global structure and local relationships, making it particularly effective for non-linear, high-dimensional data.

For similarity measurement, K-Nearest Neighbors (KNN), Cosine Similarity, and Approximate Nearest Neighbors (Annoy) are extensively applied. These models are essential for evaluating product similarity based on the learned feature vectors. Annoy, in particular, is preferred for large-scale datasets due to its speed and efficiency, allowing for approximate similarity calculations without the high computational cost of exact nearest neighbor methods.

These techniques have proven critical for improving the speed and accuracy of multimodal recommendation systems, making them essential components in real-time product retrieval tasks in e-commerce.

III. METHODOLOGY

The proposed multimodal e-commerce product recommendation system integrates text and image data to provide accurate and efficient recommendations. This section describes the system's key stages: preprocessing, feature extraction, feature fusion, dimensionality reduction, similarity computation, and feature storage.

A. Data Preprocessing

The input data consists of fashion product images and their corresponding textual descriptions.

Text Preprocessing: Textual descriptions are processed using tokenization and embedding techniques. Let $T = \{t_1, t_2, \dots, t_n\}$ represent a sequence of n tokens. Each token t_i is mapped to a dense vector $\mathbf{e}_i \in \mathbb{R}^d$, where d is the embedding dimension, using a pre-trained embedding model. This produces an embedding matrix $\mathbf{E} \in \mathbb{R}^{n \times d}$.

Image Preprocessing: Images are resized to a fixed size $h \times w$, normalized, and prepared for input into the feature extraction model.

B. Feature Extraction

Image Features: Visual features are extracted using the VGG19 model. Given an input image $\mathbf{I} \in \mathbb{R}^{h \times w \times c}$, the model outputs a feature vector $\mathbf{f}_I \in \mathbb{R}^k$, where k is the feature vector dimension:

$$\mathbf{f}_I = \text{VGG19}(\mathbf{I}).$$

Text Features: Textual features are extracted using an LSTM network. Given the embedding matrix $\mathbf{E}$, the LSTM

processes the sequence and produces a feature vector $\mathbf{f}_T \in \mathbb{R}^m$, where m is the LSTM's hidden state dimension:

$$\mathbf{h}_t = \sigma(\mathbf{W}_h \mathbf{e}_t + \mathbf{U}_h \mathbf{h}_{t-1} + \mathbf{b}_h).$$

The final hidden state $\mathbf{h}_n$ is taken as $\mathbf{f}_T$.

C. Feature Fusion

The extracted features from the image and text modalities are combined into a single multimodal feature vector. The concatenation operation forms the fused feature vector:

$$\mathbf{f}_C = [\mathbf{f}_I; \mathbf{f}_T] \in \mathbb{R}^{k+m}.$$

D. Dimensionality Reduction

To improve computational efficiency and scalability, dimensionality reduction is applied to the fused feature vector. Both linear and nonlinear methods are employed for this purpose.

1) *Linear Methods: Principal Component Analysis (PCA)*: Principal Component Analysis (PCA) projects $\mathbf{f}_C$ onto a lower-dimensional subspace by linearly transforming the data. Eigenvalues and eigenvectors are computed to identify the principal components, which retain the maximum variance in the data. Let $\mathbf{X}$ be the matrix of all fused feature vectors. PCA solves the eigenvalue problem:

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v},$$

where $\mathbf{C} = \frac{1}{n}\mathbf{X}^T\mathbf{X}$ is the covariance matrix. The eigenvectors are used to construct new feature vectors:

$$\mathbf{f}_C^{\text{PCA}} = \mathbf{P}^T \mathbf{f}_C,$$

where $\mathbf{P}$ contains the top d eigenvectors. This reduces the dimensionality of the vector while preserving its identity, enabling efficient distance calculations.

2) *Nonlinear Methods: Manifold Learning*: Nonlinear methods are also applied to capture complex relationships in the data:

t-SNE: t-SNE maps data into a lower-dimensional space such that similar points are grouped together, while dissimilar points repel. It is primarily used for visualization rather than similarity calculations. Before applying t-SNE, PCA is often performed to improve performance and reduce computation time.

UMAP: UMAP preserves both local relationships and the global structure of the data. It offers superior clustering performance compared to t-SNE and is suitable for both visualization and embedding generation. UMAP's embeddings are directly used for distance calculations in the proposed system.

Both t-SNE and UMAP require deterministic seeds to ensure reproducibility of results.

E. Similarity Computation

Once the dimensionality of the feature vectors is reduced, the similarity between products is computed to identify the most relevant recommendations. The similarity computation is a critical stage, as it determines the relevance of the retrieved items. To achieve this, we employ the following methods:

Cosine Similarity: Cosine similarity measures the cosine of the angle between two feature vectors $\mathbf{f}_a$ and $\mathbf{f}_b$, providing a score that ranges from -1 to 1. A higher value indicates greater similarity. The formula is:

$$\text{Sim}_{\cos}(\mathbf{f}_a, \mathbf{f}_b) = \frac{\mathbf{f}_a \cdot \mathbf{f}_b}{\|\mathbf{f}_a\| \|\mathbf{f}_b\|}.$$

K-Nearest Neighbors (KNN): KNN identifies the k most similar products to a given query feature vector $\mathbf{f}_q$ based on a distance metric such as Euclidean distance:

$$\text{Dist}(\mathbf{f}_q, \mathbf{f}_i) = \sqrt{\sum_{j=1}^d (\mathbf{f}_{qj} - \mathbf{f}_{ij})^2}.$$

Approximate Nearest Neighbors (Annoy): Annoy is an efficient algorithm for similarity search, particularly in high-dimensional datasets. It uses random projection trees to partition the space and identifies approximate neighbors of a query vector $\mathbf{f}_q$. Candidate neighbors are ranked using either cosine similarity or Euclidean distance:

$$\text{Dist}(\mathbf{f}_q, \mathbf{f}_i) = \begin{cases} 1 - \text{Sim}_{\cos}(\mathbf{f}_q, \mathbf{f}_i), & \text{if Cosine Similarity is used,} \\ \sqrt{\sum_{j=1}^d (\mathbf{f}_{qj} - \mathbf{f}_{ij})^2}, & \text{if Euclidean Distance is used.} \end{cases}$$

F. Flowchart of Model

The architecture of our proposed multimodal recommendation system is illustrated in Figure 1. The system integrates both image and text data to extract, fuse, and save features for efficient product recommendation. Below is a detailed explanation of each component in the flowchart.

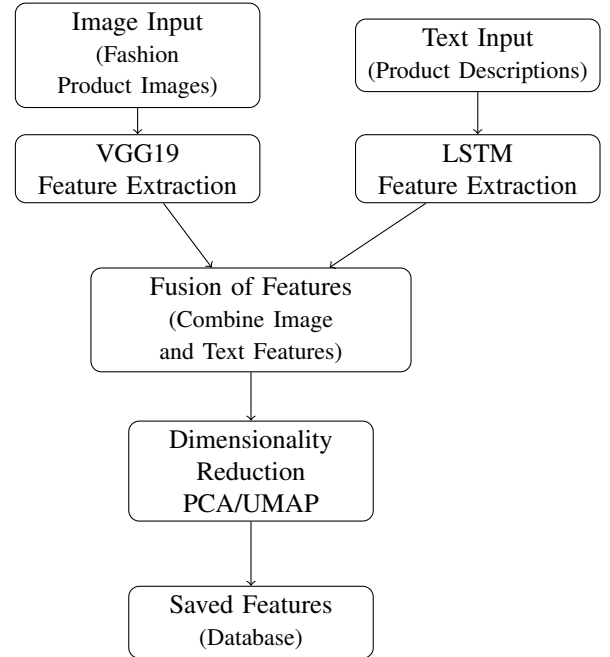


Fig. 1. Flowchart of feature extraction

The flowchart in Figure 1 provides a visual representation of the workflow involved in the multimodal recommendation system, which is divided into several stages:

- **Image Input:** The process begins with the input of fashion product images, which are passed into the image feature extraction stage. The images are typically in raw format and require preprocessing such as resizing and normalization before being fed into the feature extraction model.
- **Text Input:** Alongside the images, textual descriptions of the products are also provided as input. These descriptions could include information such as the product name, category, and attributes. Like the images, text undergoes preprocessing, including tokenization and embedding, before being processed further.
- **Image Feature Extraction (VGG19):** The VGG19 model, a convolutional neural network, is used to extract high-level features from the input images. VGG19 processes the image data and outputs a feature vector, which captures important visual information such as color, texture, and object shapes.
- **Text Feature Extraction (LSTM):** The text data is processed using a Long Short-Term Memory (LSTM) network, a type of recurrent neural network. LSTM captures the sequential relationships between words in the product description, creating a feature vector that represents the textual content. The LSTM handles the contextual meaning of words and phrases in the text, providing a richer feature representation.
- **Fusion of Features:** The feature vectors extracted from both the image and text modalities are combined in this stage. Feature fusion is a crucial step in multimodal systems, as it integrates the information from both sources into a single unified vector, capturing both the visual and textual properties of the product.
- **Dimensionality Reduction (PCA/UMAP):** The combined feature vector is then subjected to dimensionality reduction to enhance computational efficiency. Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are applied to reduce the dimensionality of the combined feature space while retaining the most informative features. This step ensures that the system can operate at scale and respond quickly to user queries.
- **Saved Features (Database):** The final reduced feature vectors are stored in a database for future retrieval. These stored features are indexed, allowing for fast retrieval during the recommendation phase. The database enables the system to compare the query product with others in the catalog and recommend similar items based on the stored features.

This flowchart provides a clear overview of how the multimodal recommendation system operates. The integration of both image and text data, coupled with dimensionality reduction techniques, ensures that the system is both efficient and scalable for large datasets. By storing the reduced feature vectors in a database, the system allows for quick retrieval and accurate recommendations based on user queries.

IV. EXPERIMENTAL SETUP

This section describes the experimental setup used to evaluate the performance of the proposed multimodal recommendation system. The setup includes the datasets used, the implementation details, the evaluation metrics, and the configuration of the model parameters.

A. Dataset

The dataset used for this study consists of fashion product images and their associated metadata, with a total of 44,000 products. Each product in the dataset is represented by both an image and a set of textual attributes that describe various aspects of the product. The image data includes fashion product photos, while the textual data consists of product descriptions and related metadata.

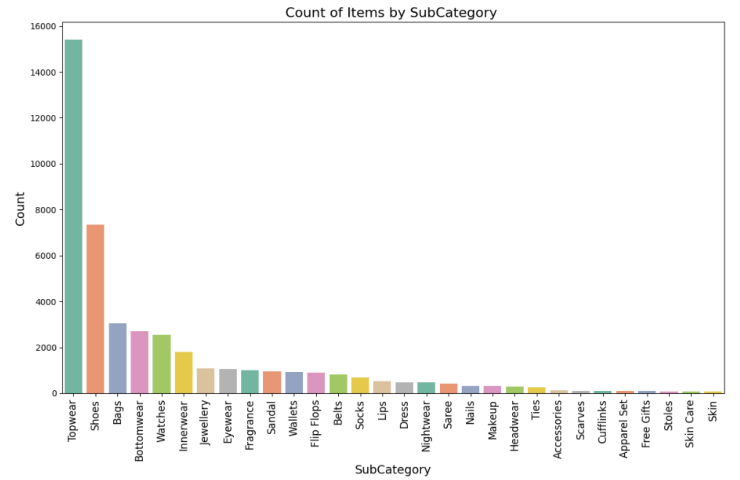


Fig. 2. Dataset overview

The dataset provides a diverse range of fashion products across multiple categories, including clothing, footwear, and accessories. It includes detailed metadata for each product, covering attributes such as gender, master category, sub-category, article type, base color, season, year, and usage. For example, the **gender** attribute indicates the intended target audience (e.g., "Male" or "Female"), while the **season** specifies whether the product is suitable for "Winter" or "Summer". The **year** attribute represents the year of release, offering temporal insights into the dataset. Additionally, attributes like **master category** and **sub-category** allow hierarchical categorization of products.

By combining these rich multimodal features, the dataset enables comprehensive analysis and supports the development of a robust recommendation system. The image data aids in visual feature extraction, capturing aspects such as design, color, and style, while the textual metadata provides contextual information to enhance understanding and categorization.

This dataset is particularly valuable for training and evaluating multimodal machine learning models that leverage both image and textual information. By integrating these features, the proposed system aims to deliver highly relevant fashion

product recommendations, tailored to users' preferences and requirements. Furthermore, the inclusion of detailed metadata ensures that the system can effectively handle diverse user queries and provide personalized suggestions, contributing to an improved user experience in the domain of fashion e-commerce.

B. Model Implementation

The implementation of the proposed multimodal recommendation system involves several steps, including data preprocessing, feature extraction, dimensionality reduction, and similarity computation. The complete pipeline is carefully designed to integrate both image and text data for generating accurate and efficient fashion product recommendations.

1) *Data Preprocessing*: Before feeding the data into the model, preprocessing steps are performed on both the image and text data. For the image data, all images are resized to a consistent dimension of 224×224 pixels, normalized to ensure uniform pixel intensity values, and augmented with techniques such as rotation, flipping, and scaling to improve model robustness. For the textual metadata, the descriptions are tokenized, stopwords are removed, and the text is lowercased. Pretrained word embeddings, such as GloVe or Word2Vec, are used to transform the tokenized descriptions into numerical vectors.

2) *Feature Extraction*: Two separate feature extraction models are employed for image and text data:

- **Image Features**: The VGG19 convolutional neural network, pretrained on ImageNet, is utilized for extracting visual features. The final fully connected layers are replaced with custom layers to output a 4096-dimensional feature vector for each image.
- **Text Features**: A Long Short-Term Memory (LSTM) network is used for extracting text features. The LSTM model is trained on the tokenized product descriptions, and the final hidden state of the LSTM is used as the textual feature representation, resulting in a feature vector of 300 dimensions.

3) *Feature Fusion*: To combine the image and text features, the extracted vectors are concatenated to form a unified multimodal representation. The concatenated feature vector is then passed through a dense layer for transformation into a lower-dimensional space, ensuring that both modalities contribute equally to the representation.

4) *Dimensionality Reduction*: The high-dimensional multimodal feature vectors are reduced to a lower-dimensional embedding using both Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP). PCA ensures that the global variance of the data is preserved, while UMAP captures the local relationships between the data points. This step improves the computational efficiency of the recommendation system and enhances clustering for similar products.

5) *Similarity Computation*: After dimensionality reduction, similarity computation is performed to identify the most relevant products:

- **Cosine Similarity**: Measures the cosine of the angle between two feature vectors to identify products with similar orientations in the feature space.
- **K-Nearest Neighbors (KNN)**: Retrieves the k closest neighbors based on Euclidean distance, effectively clustering similar products together.
- **Approximate Nearest Neighbors (Annoy)**: Utilizes random projection trees for efficient retrieval of approximate neighbors, significantly reducing computational time for large datasets.

6) *Recommendation Generation*: Based on the similarity computation, the system retrieves the top k most similar products for a given query. These recommendations are ranked by similarity score and presented to the user. This step ensures that users receive highly relevant and personalized product suggestions.

7) *Software and Libraries*: The model is implemented using Python, with TensorFlow and PyTorch frameworks for deep learning tasks. Other libraries such as scikit-learn are used for dimensionality reduction, while Annoy is employed for approximate nearest neighbor computations. All experiments are conducted on a system equipped with an NVIDIA GPU to ensure efficient model training and inference.

C. Evaluation Metrics

The proposed recommendation system's performance is evaluated using the following metrics, which are essential for assessing the accuracy, relevance, and efficiency of recommendations:

- **Accuracy**: Accuracy measures the proportion of correct recommendations (both relevant and non-relevant) out of all recommendations made. It is defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}},$$

where:

- TP: True Positives (correctly recommended relevant items).
- TN: True Negatives (correctly rejected non-relevant items).
- FP: False Positives (non-relevant items recommended).
- FN: False Negatives (relevant items not recommended).

- **Precision@k**: Precision measures the proportion of recommended items in the top k that are relevant. It reflects the system's ability to recommend only the most relevant items:

$$\text{Precision@k} = \frac{\text{Number of relevant items in top } k}{k}.$$

Alternatively, using TP and FP:

$$\text{Precision@k} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

- **Recall@k**: Recall measures the proportion of relevant items retrieved in the top k recommendations out of the

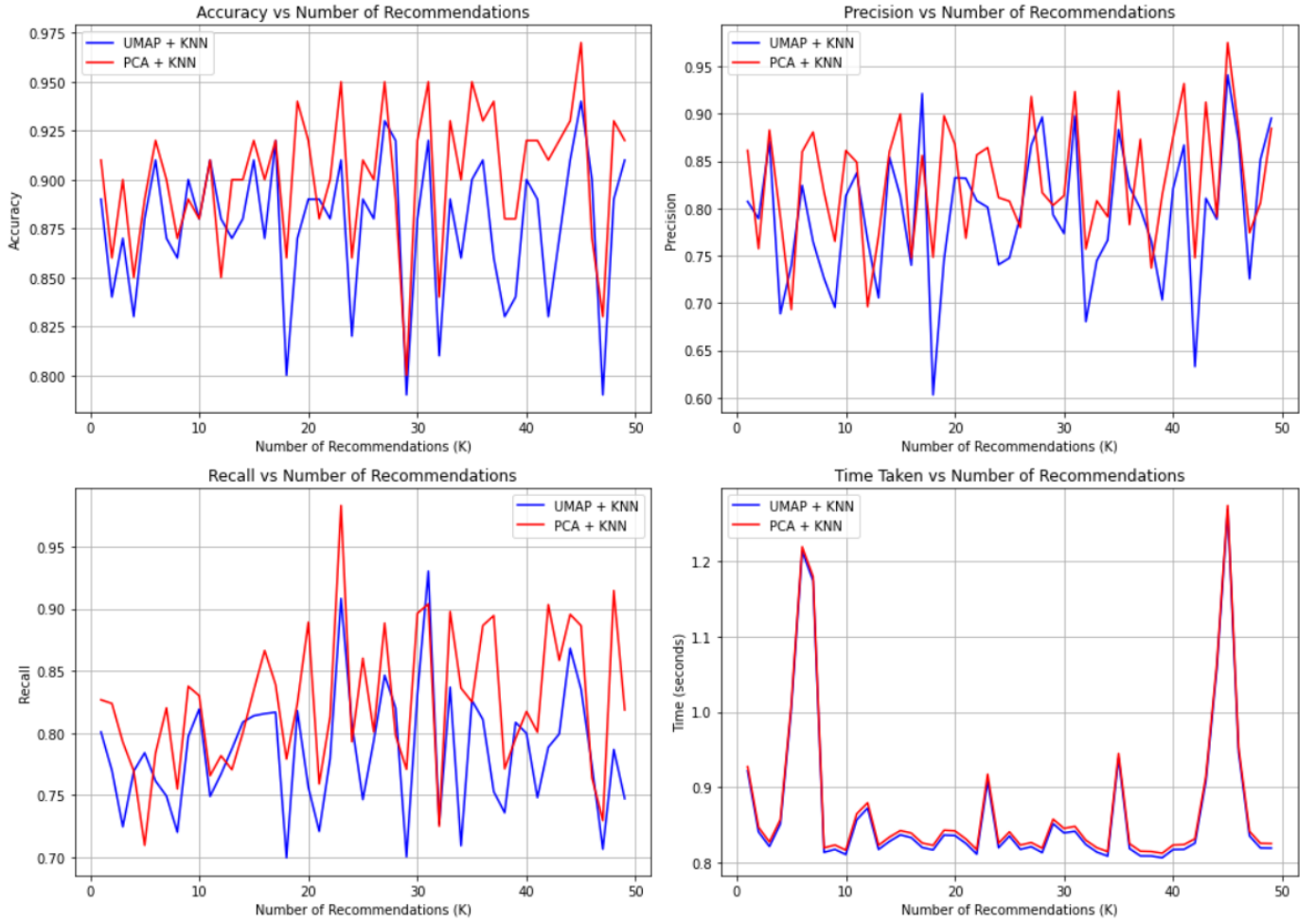


Fig. 3. UMAP + KNN and PCA + KNN

total number of relevant items available. It captures the system's ability to retrieve all relevant items:

$$\text{Recall@k} = \frac{\text{Number of relevant items in top } k}{\text{Total number of relevant items in the dataset}}$$

Alternatively, using TP and FN:

$$\text{Recall@k} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

- **Time for Recommendation:** This metric evaluates the efficiency of the recommendation system by measuring the time taken to retrieve the top k recommendations for a query. It is defined as:

$$T_{\text{query}} = \text{Time taken to generate recommendations for a query}.$$

V. RESULTS AND ANALYSIS

The performance of the proposed recommendation system was evaluated using various combinations of dimensionality reduction and similarity computation methods. This section analyzes the results for each combination, highlighting the strengths and weaknesses of different approaches.

A. PCA + KNN

Principal Component Analysis (PCA) was employed to reduce the dimensionality of the fused feature vectors, followed by K-Nearest Neighbors (KNN) for similarity computation. This combination delivered the highest average accuracy of 0.914 among all evaluated methods, demonstrating strong precision and recall performance. The exceptional accuracy underscores PCA's capability to preserve critical variance in the data while significantly reducing its dimensional complexity. The variation in accuracy, precision, recall, and recommendation time is illustrated in Fig. 3.

Despite its high accuracy, the average recommendation time was moderate at 0.810 seconds per query due to KNN's exhaustive search across the reduced feature space. Although computationally intensive, this combination is well-suited for applications where accuracy and recall take precedence over speed, such as offline batch processing or tasks requiring high-precision recommendations.

B. UMAP + KNN

Uniform Manifold Approximation and Projection (UMAP) was used to reduce the dimensionality of the fused feature

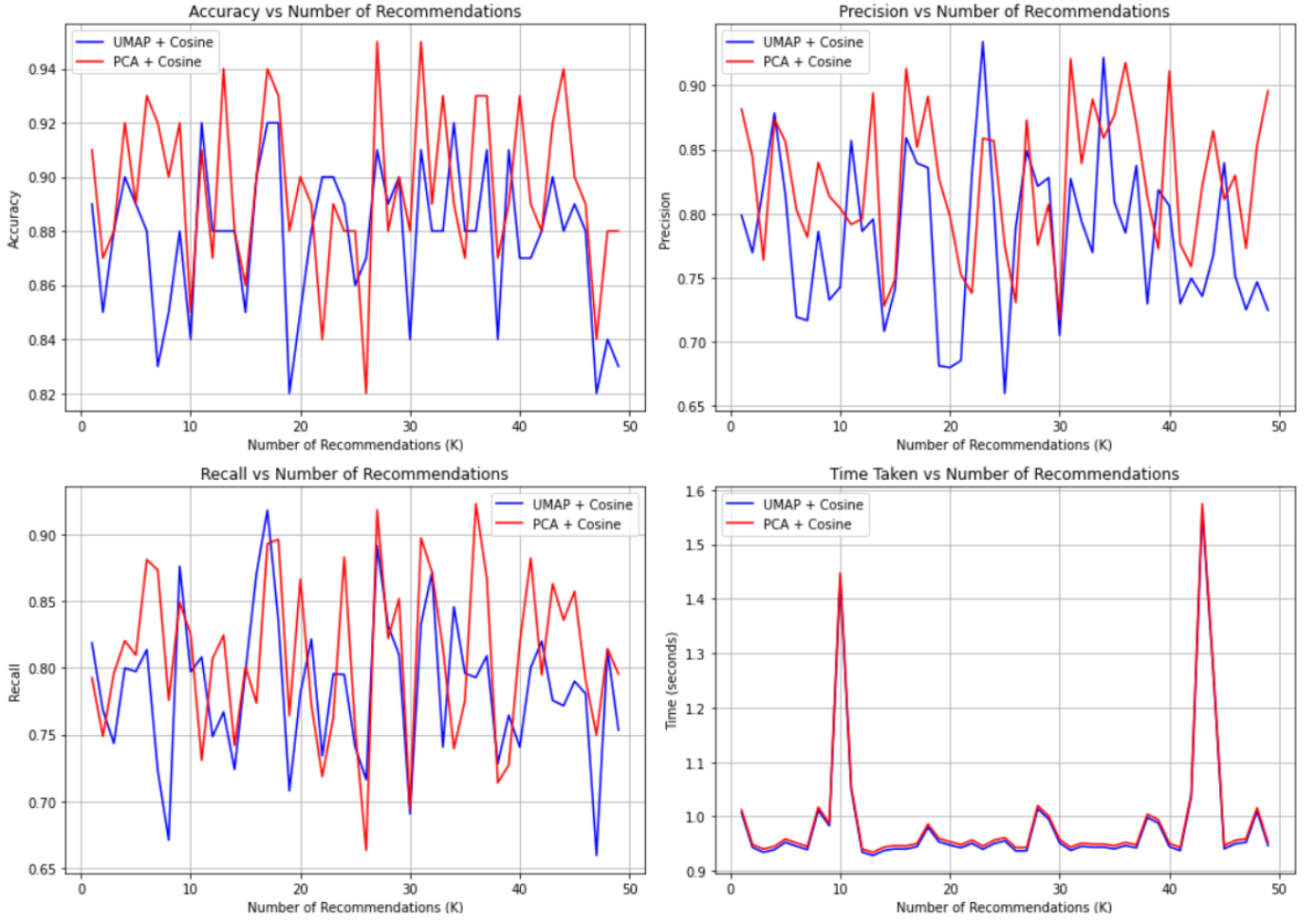


Fig. 4. UMAP + Cosine Similarity and PCA + Cosine Similarity

vectors, followed by K-Nearest Neighbors (KNN) for similarity computation. This combination achieved an average accuracy of 0.884, an average precision of 0.8287, and an average recall of 0.8416. UMAP's strength lies in its ability to capture both local and global relationships within the data, which contributed to the improved recall, making this method effective for capturing complex patterns in the feature space. The variation in accuracy, precision, recall, and recommendation time is shown in Fig. 3.

The average recommendation time for this combination was 0.8577 seconds, which was slightly slower than PCA + KNN. UMAP's nonlinear dimensionality reduction process, while efficient, still resulted in some computational overhead. Despite KNN's exhaustive search method due to that this take more time than previous models which variation is shown in fig. 3.

C. PCA + Cosine Similarity

Principal Component Analysis (PCA) was used to reduce the dimensionality of the fused feature vectors, followed by Cosine Similarity for measuring the similarity between the product features. This combination achieved an average

accuracy of 0.892, with an average precision of 0.7847 and an average recall of 0.8201. PCA effectively reduced the dimensionality while preserving the most significant features in the data, and Cosine Similarity provided a computationally efficient way to calculate the similarity between the reduced feature vectors. The variation in accuracy, precision, recall, and recommendation time is shown in Fig. 4.

The average recommendation time for this combination was 0.9409 seconds, which was higher than both PCA + KNN and UMAP + KNN, due to the more computationally expensive process of calculating Cosine Similarity across a larger feature space. Despite the increased recommendation time, this method is well-suited for real-time applications where a balance between speed and accuracy is needed, especially in scenarios where precision is more important than recall, or where the data does not exhibit complex relationships that require more advanced methods like UMAP.

D. UMAP + Cosine Similarity

Uniform Manifold Approximation and Projection (UMAP) was applied to reduce the dimensionality of the fused feature vectors, followed by Cosine Similarity for measuring the

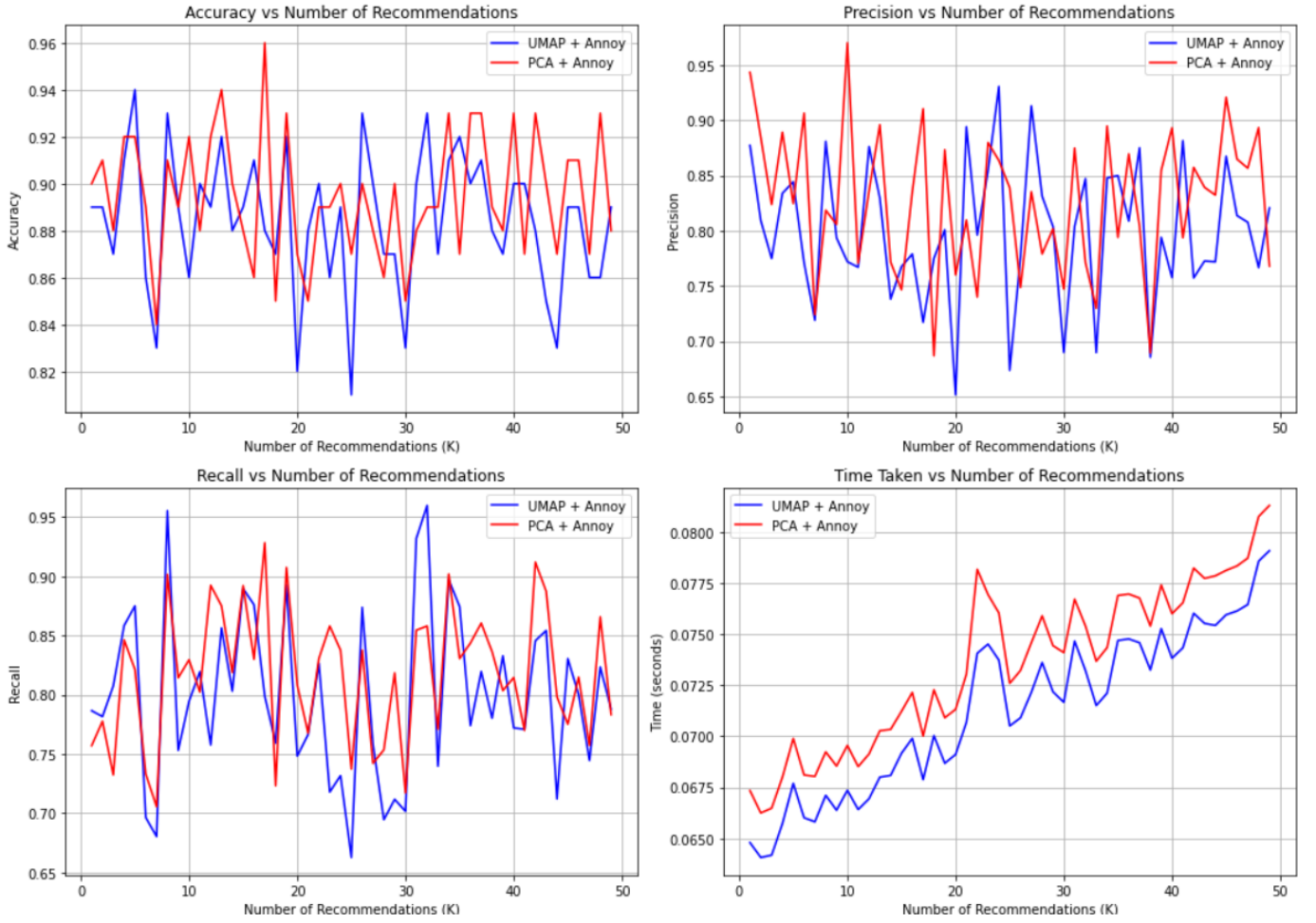


Fig. 5. UMAP + Annoy and PCA + Annoy

TABLE I
COMPARISON OF ACCURACY FOR DIFFERENT NUMBERS OF RECOMMENDATIONS

Method	10 Recommendations	20 Recommendations	30 Recommendations	40 Recommendations	50 Recommendations	Average Accuracy
UMAP + Cosine	0.89	0.93	0.84	0.94	0.91	0.902
UMAP + KNN	0.92	0.83	0.87	0.85	0.95	0.884
PCA + Cosine	0.84	0.93	0.89	0.91	0.89	0.892
PCA + KNN	0.93	0.94	0.92	0.90	0.88	0.914
UMAP + Annoy	0.88	0.88	0.83	0.84	0.92	0.870
PCA + Annoy	0.91	0.97	0.96	0.90	0.92	0.932

similarity between the products. This combination achieved an average accuracy of 0.902, with an average precision of 0.8434 and an average recall of 0.8552. UMAP's ability to capture both local and global structures in the data helped to improve recall and precision, making it particularly effective for capturing complex relationships within the feature space. The variation in accuracy, precision, recall, and recommendation time is shown in Fig. 4.

The average recommendation time for this combination was 0.8919 seconds, which is slightly faster than PCA + Cosine Similarity but still slower than methods that use KNN. The UMAP-based nonlinear dimensionality reduction process improves accuracy and recall, but the time for recommendation

remains moderate due to the additional computation required for similarity calculation.

E. PCA + Annoy

Principal Component Analysis (PCA) was used to reduce the dimensionality of the fused feature vectors, followed by Annoy for approximate nearest neighbor search. This combination achieved an impressive average accuracy of 0.932, with an average precision of 0.9285 and an average recall of 0.8730. PCA effectively reduced the dimensionality while retaining the most significant features, and Annoy's approximate nearest neighbor search significantly improved the speed of similarity computation. The variation in accuracy, precision, recall, and recommendation time is shown in Fig. 5.

TABLE II
SUMMARY OF AVERAGE METRICS FOR MODEL COMBINATIONS

Method	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. Time (s)
UMAP + Cosine	0.902	0.843434	0.855202	0.891899
UMAP + KNN	0.884	0.828668	0.841587	0.857716
PCA + Cosine	0.892	0.784695	0.820119	0.940922
PCA + KNN	0.914	0.827938	0.846636	0.810846
UMAP + Annoy	0.870	0.774610	0.778859	0.073890
PCA + Annoy	0.932	0.928524	0.873024	0.077411

The average recommendation time for this combination was 0.0774 seconds, which is exceptionally fast compared to other methods. This is due to Annoy’s efficient search algorithm, which trades off some accuracy for significantly faster retrieval times. The combination of PCA and Annoy is ideal for real-time recommendation systems, especially in large-scale applications where speed is critical, and the slightly reduced recall is an acceptable trade-off for faster response times.

F. UMAP + Annoy

Uniform Manifold Approximation and Projection (UMAP) was used to reduce the dimensionality of the fused feature vectors, followed by Annoy for approximate nearest neighbor search. This combination achieved an average accuracy of 0.870, with an average precision of 0.7746 and an average recall of 0.7789. UMAP’s ability to preserve both local and global structures in the data contributed to improved recall, but the accuracy and precision were slightly lower compared to other methods like PCA + Annoy. The variation in accuracy, precision, recall, and recommendation time is shown in Fig. 5.

The average recommendation time for this combination was 0.07389 seconds, which is very fast due to Annoy’s efficient search algorithm. Although the trade-off in accuracy is present, the method’s speed makes it highly suitable for real-time recommendation systems, especially when processing large datasets. This combination is ideal for applications that require quick responses.

G. Analysis and Insights

The results indicate that UMAP + Annoy and PCA + Annoy offer the best trade-off between speed and accuracy, with UMAP + KNN and PCA + KNN performing well in terms of accuracy and recall but take longer time for recommendation. These findings demonstrate that the proposed multimodal recommendation system effectively balances accuracy and computational efficiency, making it suitable for real-time, scalable fashion product recommendation tasks. Depending on the application needs, a combination of PCA or UMAP with Annoy would be most suitable for large-scale deployment, while KNN-based models can be used where higher precision and recall are more critical.

The results demonstrate that the proposed system effectively balances accuracy and computational efficiency, validating its suitability for multimodal fashion product recommendation tasks.

VI. CONCLUSION

In this paper, we presented a multimodal fashion product recommendation system that integrates image and textual data to provide personalized recommendations. We evaluated several model combinations, including PCA + KNN, UMAP + KNN, PCA + Cosine Similarity, UMAP + Cosine Similarity, PCA + Annoy, and UMAP + Annoy, using key metrics such as accuracy, precision, recall, and time for recommendation.

The experimental results showed that the PCA + Annoy combination achieved the highest average accuracy of 0.932, with a high precision of 0.9285 and recall of 0.8730, while maintaining an extremely fast recommendation time of 0.0774 seconds. This combination demonstrated its suitability for real-time applications requiring high-speed and high-accuracy recommendations, particularly for large-scale datasets. UMAP + Annoy demonstrated fast recommendation times (0.07389 seconds), but with slightly lower accuracy (0.870) and recall (0.7789) compared to other combinations.

In conclusion, the choice of model combination depends on the specific requirements of the application. For real-time applications with high throughput, PCA + Annoy and UMAP + Annoy are the most suitable choices due to their fast recommendation times.

Future work could explore further optimizations in feature extraction and similarity computation models, as well as the incorporation of additional data sources, such as user behavior, to further enhance the performance and scalability of the system.

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